

Algebra II with Trigonometry

Chapter 6.4

Olympiad Problems (Gemini)

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Problems

1. Suppose x is an angle such that $\sec(2024x) + \tan(2024x) = 7$. Find the exact value of $\sec(2024x) - \tan(2024x)$.
2. Compute the exact value of $\tan\left(\frac{\pi}{12}\right) + \cot\left(\frac{\pi}{12}\right)$.
3. Evaluate the sum of all x -values strictly between 0 and 2π where the graph of $y = \sec(8x)$ has a vertical asymptote.
4. Determine the sum of the first ten positive x -intercepts of the graph of the function $f(x) = \cot\left(\pi x - \frac{\pi}{4}\right)$.
5. Find the smallest positive real number x that satisfies the equation $\sec\left(3x - \frac{\pi}{6}\right) = \csc\left(3x - \frac{\pi}{6}\right)$.
6. Determine the fundamental period of the function $f(x) = |\tan x| + |\cot x|$.